

# AI-Driven Water Quality Monitoring and Remediation System

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**Abstract**—This paper talks about an AI-powered system that is meant to change the way water quality monitoring works from just gathering raw data to making useful information. Conventional systems usually provide sensor readings without context, such as turbidity and pH, which results in postponed laboratory analysis for interpretation. The suggested system combines various sensors with an AI interpretation layer to categorize water quality and produce remediation advisories in real time. An Action Priority Index (API) for automated environmental recommendations and an understandable dashboard for temporal trend analysis are two important innovations. Experimental results demonstrate high classification accuracy and efficient contextual reasoning for various water contamination scenarios.

## I. INTRODUCTION

The inability to interpret sensor data instantly is the current problem in water management. The majority of current systems function as basic data loggers, creating a bottleneck where manual intervention is required for contamination analysis. This project presents an AI-powered solution that explains sensor data using natural language and contextual reasoning. Through the use of an Action Priority Index, the system not only checks for the quality of the water but also recommends quick fixes for cleanup of contaminated water bodies.

The proposed system aims to bridge the gap between raw sensor measurements and decision-making by incorporating an Action Priority Index (API), which quantifies the urgency of intervention based on contamination severity. Unlike traditional monitoring systems that simply display numerical values, this approach provides a structured interpretation layer capable of categorizing water quality conditions and recommending suitable remediation actions. Moreover, the combined use of turbidity sensor and the color detection improves the effectiveness of contamination measurement by taking into consideration the particle's concentration and characteristic features. This system also let us continuously monitor through a dashboard or an interference, to let the user to observe temporal trends so as to make the necessary decisions.

This approach demonstrates how embedded sensors concepts along with AI-based reasoning can effectively change the traditional monitoring into an intelligent advisory system, contributing toward efficient resource management and improved public health safety.

## II. LITERATURE SURVEY

The convergence of the Internet of Things (IoT) with edge computing and artificial intelligence (AI) has advanced the technological trend of water quality monitoring. A previous study from [1] reviewed an IoT based Aquaponics Facility with a Data Acquisition/Monitoring Dashboard, demonstrating that Multi-Sensors can be utilized with Real-Time Dashboards to continuously monitor Environmental Parameters. Olisa et al. [5] developed a Smart Two-Tank System designed to Monitor Water Quality and Water Level using a Micro-Controller with Sense Nodes connected to Cloud Data Upload Provisioning for Real-Time Environmental Parameters.

Recent literature highlights the importance of Edge Processing as a fundamental principle behind ESP32 based Architecture. Correa et al. [2] proposed an Edge Processor integrated into an Underground Water Body Monitoring System, showing that Local Computation on Less Capable Hardware reduces Time Delay and Reliance upon Cloud Computing. Ren et al. [4] evaluated Edge Computing Methods enabling faster Anomaly Detection and reduced Transmission Overhead. Kumar and Udgata [10] discussed installing Machine Learning Models onto IoT Edge Devices, demonstrating that Lightweight Classifying Schemes can function within Hardware Deficient Environments.

Machine learning and AI approaches transform raw sensor readings into beneficial quality assessments. Rahu et al. [8] proposed water-quality forecasting using IoTs with linear regression and decision tree classifier methods on multi-sensor data for contamination prediction. Roy et al. [9] utilized IoT and machine learning regression for measuring water quality. Chhipi-Shrestha et al. [6] explored AI and Soft Computing approaches including Fuzzy Logic and Deep Neural Networks (DNN) for categorizing contamination levels. Research work [3] presented an Intelligent IoT System for Real-Time Water Quality and Pollution Detection using Multiple Sensors with intelligent back-end processing, while Durgun [7] utilized AI-Powered Urban Water Systems to identify contaminants and provide Advisory with Low Latency.

A large body of literature exists regarding Sensing, Edge Computing and Artificial Intelligence. However, integration of Sensor Acquisition, Edge-Level Classification and Artificial Intelligence Contextual Reasoning into a Dashboard that explains how to fix problems remains limited. To achieve

this objective, a Four-Layered System is developed where the ESP32 performs Real-Time classification, a Python Intelligence Back End analyzes Trends Over Time, and a Live Web-Based Dashboard displays Advice in an understandable manner. These systems combine Sensing, Edge Computing and Artificial Intelligence into a Single Pipeline.

### III. METHODOLOGY

#### A. Architecture with four layers

Architecture with Four Layers: Layer number 1: Acquisition of Data/Sensing: TCS3200 color sensor is used to identify chemical discharge, presence of sediments or dyes and a turbidity sensor to measure suspended particles. Layer number 2: Real time data processing (ESP32 board): Measures the Action Priority Index, formats the raw readings into JSON format, finds the water status (Clean, Moderate, and Heavy), and converts sensor data from analog into digital values. Layer number 3: Interpretation of data using AI (Python): Produces human-readable explanations and correct actions while monitoring quality patterns (stable, improving, or degrading). Layer number 4: Visualization (Dashboard): Offers an instantaneous decision support user interface and real-time metrics display.

#### B. Hardware Requirements

- Microcontroller used: ESP32 Development Board
- Sensors: Turbidity Sensor, TCS3200 Color Sensor
- Indicators and Setup: Jumper Wires, Breadboard, USB Cable, Container

#### C. Software tools

- Sensor data: Arduino IDE, C++
- Backend and Processing: Python 3, PySerial, Flask (creates local server)
- Frontend Dashboard: HTML, CSS, JavaScript
- AI/Logic: Local LLM / AI Advisory logic

#### D. Analytical Description

##### 1. Signal Processing and Normalization process:

- The 12-bit ADC of the ESP32 board is used, which captures the analog values from 0 to 4095.
- Turbidity: Raw values are mathematically normalized to a 0-100 percent scale for human-readable dashboard display.

##### 2. Logic of classification or Thresholds: Based on the calibrated raw turbidity values (T raw):

- Clean: T-row is less than 1200
- Moderate: 1200 is less than or equal to T-row greater than or equal to 2000
- Heavy: T-row is greater than 2000

##### 3. Action Priority Index (API) Formulation

- API is a computed theoretical metric scaled between 0 and 1.
- It functions as a weighted sum of turbidity percentage and color deviation (identifying abnormal RGB shifts indicating dyes or algae).

##### 4. AI Decision Matrix

- The Python code logic makes use of the decision matrix which is ultimately based on API + Temporal Trend (Stable, Improving, Degrading).
- For example: If the API displayed is  $\leq 0.7$  AND the trend is degrading, then the AI outputs high-urgency remediation (e.g., immediate filtration).

#### E. Software Algorithm/Workflow:

- The proposed system follows a defined flowchart for gathering and processing data:
- In the start, the ESP32 board is set creating a stable connection between the sensors and the software.
- The turbidity sensor detects the amount of solute particles, whereas the TCS3200 measures and calculates the RGB values to figure out the particular kinds of contaminants, such as dyes, algae, and sediment.
- The data is digitalized i.e. the ESP32's ADC converts the raw sensor data into a standardized format, representing it as percentages of impurities.
- The system integrates all this sensor data i.e. the color and turbidity readings, to decide the Action Priority Index (API), which falls within a range of 0.0 to 1.0.
- The system categorizes water samples into five levels of contamination, ranging from Clean, Slightly Concerning, Moderate, Heavy, to Dangerous.
- The metrics are compiled into a JSON package and sent to the Python environment through a serial connection as part of the data transmission process.
- The AI interpretation part includes the Python environment, analyzing the patterns and producing insightful AI commentary through deep reasoning.
- The result is sent to a web page dashboard, which displays the data outcome on it.
- This procedure of live monitoring and decision support repeats perpetually till we stop the system.

### IV. RESULTS AND ANALYSIS

This system receives the raw turbidity and color sensor readings. The raw sensor data is processed through the arduino IDE and then, this system presents information like the contamination percentage, classification based on the contamination, details about it and recommendations as well.

The turbidity content measured helps to assess the amount of contaminated particles, while color sensor values are used to determine the type of contamination, such as sediment, organic matter, or chemicals. These values are then used to set the Action Priority Index (API) indicating the urgency of the action recommended.

As the amount of impurity in the water rises, the contamination percentage and the API value show a steady increase. The findings are dependable and uniform across all five contamination levels.

#### A. Parameters Considered:

- Turbidity percentage: This indicates the level of suspended particles present in the water. RGB Values: Help

to find out the properties of impurities present in the water.

- Action Priority Index (API): This numeric value ranging from 0 to 1 indicates the urgency of the recommended action to be taken.
- Classification Level: This parameter indicates the degree of pollution present in the water.
- Visual Analysis: This factor aids in determining the kind of contaminants present in the water.
- AI Recommendation: This setting shows suggested actions.

#### B. Readings and Observations:

TABLE I  
OBSERVATION TABLE

| Condition (ap-prox.) | Observed Contamination % | Visual Analysis             | Classification      |
|----------------------|--------------------------|-----------------------------|---------------------|
| Clean water          | 5.6                      | Clean water                 | Clean               |
| Pinch Haldi          | 16.3                     | Trace contamination         | Slightly concerning |
| 1/16 tsp Haldi       | 34.5                     | Moderate murkiness          | Moderate            |
| 1/8 tsp Haldi        | 69.7                     | Suspended particles visible | Moderate            |
| 1/4 tsp Haldi        | 72.9                     | High contamination          | Heavy               |
| 1/2 tsp Haldi        | 84.3                     | Opaque samples visible      | Dangerous           |

#### C. Inference

- As the amount of contaminated particles in the water increases, the percentage of impurities and also the value of the API increase steadily.
- The use of both the sensors together enhances the ability to reliably.
- The interpretation of sensor data using AI aids in decision-making, enabling faster responses to environmental issues compared to traditional testing methods.

#### V. CONCLUSION

The proposed system shows an effective solution for real-time water quality monitoring in a cost-effective way, using a combination of turbidity, color, and AI for water quality analysis. The use of the Action Priority Index and classification system gives a significant insight into the level of impurities in water, with the use of the dashboard for better understanding of the water quality analysis. This system reduces the dependency on laboratory analysis, which can be conducted in a slow manner, by converting the sensor data into an effective recommendation for better decision-making in the environment. The proposed system can be expanded with other sensors or wirelessly for use in the real world.

#### VI. SYSTEM AND OUTPUT VISUALIZATION

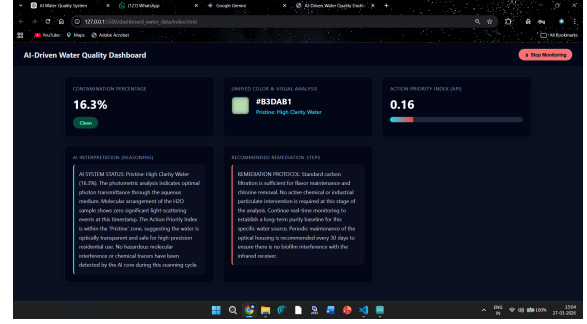


Fig. 1. Dashboard output showing contamination analysis and Action Priority Index.

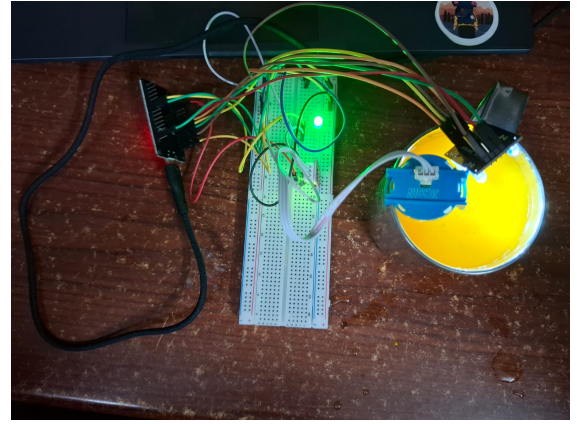


Fig. 2. Hardware setup showing ESP32, turbidity sensor, and color sensor arrangement.

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